HAGEN GRESS

Postdoctoral Associate | Microfluidics • MEMS/NEMS • Resonators

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Projects

MICRO/-NANOELECTROMECHANICAL SYSTEMS (MEMS/NEMS) RESONATORS

November 2021 - present | Boston University

- Designed, fabricated, and characterized 3D-printed MEMS resonators with integrated electrothermal actuation and piezoresistive readout
- Fabricated metallic structures on top of lithium niobate substrates to drive polymeric resonators via surface acoustic waves
- Designed a waterproof test chamber for resonance measurements under continuous fluid flow and coordinated its fabrication with the university machine shop
- Assembled, aligned, and calibrated a Michelson interferometer to optically measure nanoscale displacements at frequencies between 0.3–70 MHz
- Investigated the thermomechanical noise of NEMS resonators in viscous fluids
- Built a phase-locked loop to track resonance frequencies in gas sensing applications
- Developed an analytical model describing the delamination dynamics and resonance behavior of pressurized two-dimensional polyaramid membranes

ANTIBACTERIAL SUSCEPTIBILITY TESTING

September 2019 - October 2021 | Boston University

- Designed and fabricated PDMS-based microchannels with integrated electrodes
- Conducted low-frequency (<10 Hz) electrical measurements of resistance fluctuations across microchannels filled with bacterial suspensions
- Compared ion fluctuations of healthy bacteria and bacteria exposed to antibiotics
- Modeled the effect of bacterial nanomotion on resistance fluctuations in COMSOL

Work Experience

FRESENIUS MEDICAL CARE NORTH AMERICA | Intern

May 2023 – August 2023 | Lawrence, MA

- Created a CAD assembly of a water filter for home hemodialysis equipment
- Assembled test fixtures to measure the flow rates of peristaltic pumps

MILTENYI BIOTEC GMBH | Plastics Engineer

September 2018 – July 2019 | Bergisch Gladbach, Germany

- Designed injection-molded components for a cell sorting device using CAD software
- Created 3D models and technical drawings for cell strainers

FRAUNHOFER INSTITUTE FOR LASER TECHNOLOGY (ILT) | Research Assistant

January 2017 – March 2018 | Aachen, Germany

- Assessed the suitability of different conductive photopolymers for 3D printing
- Converted a droplet dispenser into a 3D printer for hydrogels

BARTELS MIKROTECHNIK GMBH | Intern

December 2015 – July 2016 | Dortmund, Germany

- Assembled and tested microfluidic pumps in quality control
- Designed electrode arrays and 3D-printed microfluidic structures to create a digital droplet system for optical switch actuation via electrowetting-on-dielectric (EWOD)

Education

BOSTON UNIVERSITY

2025 | Boston, MA

Ph.D., Mechanical Engineering
GPA: 3.9 / 4.0

RWTH AACHEN UNIVERSITY

2018 | Aachen, Germany

M.Sc., Mechanical Engineering
GPA: 4.0 / 4.0

RWTH AACHEN UNIVERSITY

2016 | Aachen, Germany

B.Sc., Mechanical Engineering
GPA: 3.2 / 4.0

Skills

DESIGN & SIMULATION

SolidWorks (CSWA-MD) • Solid Edge • PTC Creo • KLayout • COMSOL Multiphysics

PROGRAMMING & ANALYSIS

MATLAB • OriginLab • LabVIEW

MICROFABRICATION

Wetbench • Plasma Asher • Spin Coater • Mask Writer • Mask Aligner • Dicing Saw • Thin Film Deposition Systems • Ball Bonder • PDMS Casting • Micro-/Nanoscale 3D-Printer

LABORATORY EQUIPMENT

Ultra-High Vacuum Systems • Microfluidic Flow Controllers • Optical Interferometry • Atomic Force Microscopy • Scanning Electron Microscopy • Dynamic Light Scattering

INSTRUMENTATION

Digital Multimeters • Function Generators • Oscilloscopes • Network/Spectrum Analyzers • DAQ Boards • Pre-Amplifiers • Lock-In Amplifiers • PID Controllers • Attenuators • Filters • Power Splitters • Bias Tees